

Technical Assessment

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Assessment Overview

This assessment evaluates your ability to:

- **Demonstrate the end-to-end machine learning process**
- **Communicate technical results clearly** (explain your code and methodology)
- **Interpret findings for business stakeholders** (translate metrics into actionable insights)
- **Justify your decisions** (explain why, not just what)

Instructions

Time: 2-3 hours

Deliverable: Notebook (.html) containing:

- Well-commented code with clear explanations
- Visualizations with proper labels and titles
- Written interpretations in Markdown cells
- Business recommendations in plain language

Tools: Python, pandas, scikit-learn, matplotlib/seaborn

Important: *We value your thought process and communication as much as correct code. Explain your reasoning throughout.*

Business Context

You are a data scientist at a telecommunications company. The marketing team has noticed an increase in customer churn and wants to understand:

1. Which factors are driving customers to leave?
2. Can we predict which customers are at risk of churning?
3. What actions can we take to improve customer retention?

Your task is to build a Logistic Regression model, interpret the results, and present findings that both technical and non-technical stakeholders can understand.

Dataset

Source: aai510-group1/telco-customer-churn (Hugging Face)

```
from datasets import load_dataset ds = load_dataset("aai510-group1/telco-customer-churn")
```

The dataset contains 33 features describing customer demographics, services, and billing. The target variable is **Churn** (0 = Stayed, 1 = Left).

Variable	Description
Churn	Target: 0 = Stayed, 1 = Churned
Satisfaction Score	Customer satisfaction rating (1-5)
Number of Referrals	Referrals made by customer
Online Security	Has online security (0/1)
Dependents	Has dependents (0/1)
Contract	Month-to-Month, One Year, Two Year
Referred a Friend	Has referred a friend (0/1)
Tenure in Months	Months with company
Monthly Charge	Monthly charge amount
Senior Citizen	Is senior citizen (0/1)
Premium Tech Support	Has premium support (0/1)
Streaming Music	Has streaming music (0/1)
Paperless Billing	Uses paperless billing (0/1)
Online Backup	Has online backup (0/1)
Phone Service	Has phone service (0/1)
Gender	Customer gender (encoded)
Internet Service	Has internet service (0/1)
Population	Zip code population
Internet Type	DSL, Fiber Optic, Cable
Multiple Lines	Has multiple lines (0/1)
Streaming Movies	Has streaming movies (0/1)
Avg Monthly GB Download	Avg monthly GB downloaded
Under 30	Is under 30 years old (0/1)
Offer	Marketing offer accepted
Payment Method	Bank, Credit Card, Mailed Check
Device Protection Plan	Has device protection (0/1)
Number of Dependents	Count of dependents
Partner	Has partner (0/1)
Married	Is married (0/1)
Total Charges	Total charges to date
Unlimited Data	Has unlimited data (0/1)
Avg Monthly Long Distance Charges	Avg long distance charges
Streaming TV	Has streaming TV (0/1)
Age	Customer age in years

Assessment Questions

1. Data Exploration & Understanding (20 points)

Goal: Demonstrate your ability to explore data and communicate initial findings.

- a) Load the dataset and report the size of training, validation, and test splits. What do you observe about the data structure?

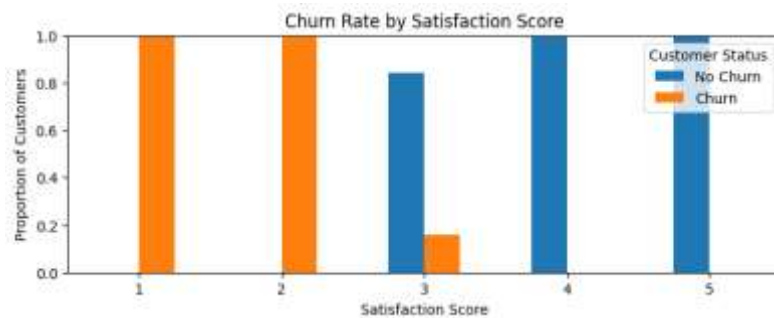
The dataset consists of a 60-20-20 split of training, validation, and test split. This is a common set-up for datasets with less than 10,000 records, as this one has around 7,000 rows overall. This data structure also involves a lot of categorical and numerical features amounting to 52 columns.

- b) Calculate the churn rate. Is this dataset balanced or imbalanced? **Explain why this matters for model training.**

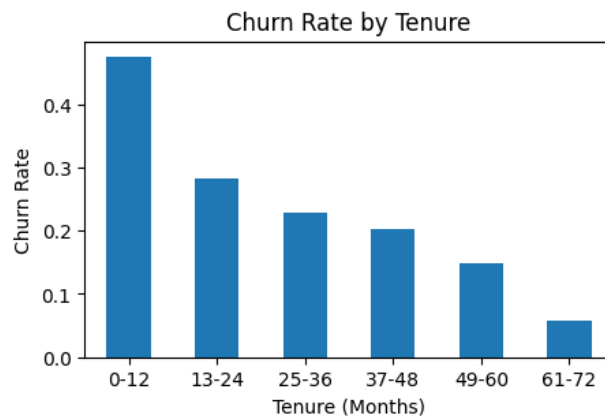
The churn rate is 26%, and this is equivalent to all three models. Though the 74-26

rate indicates an imbalance, it is still relatively balanced, allowing it to be usable. A balanced dataset is really important in classification problems since it allows the model to learn multiple classes, rather than focus on the majority class only.

- c) Create at least TWO visualizations that reveal interesting patterns related to churn (e.g., churn by contract type, churn by tenure). **Write 2-3 sentences interpreting each visualization.**

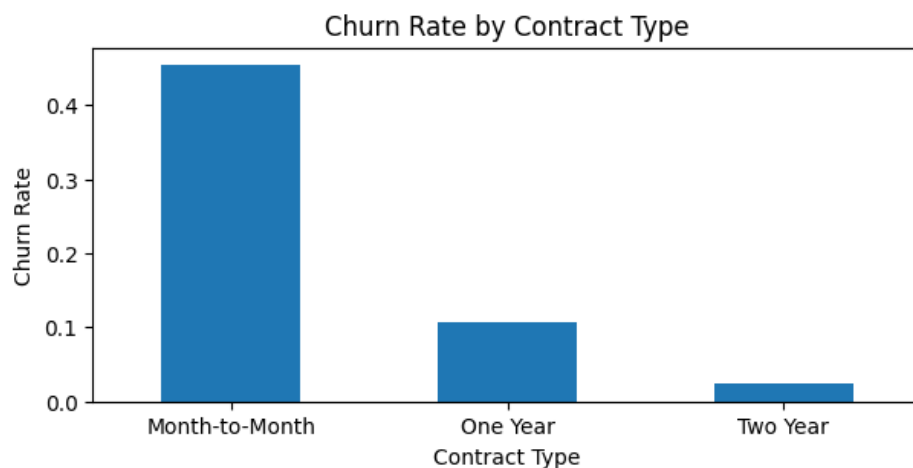


Here is the Satisfaction Score graphed with Churn. It is observed that most customers who churned are those with a satisfactory score of 1-3, while those that didn't were on 3-5.



Here is the churn rate as the time of service went on, using Tenure in Months. From here, we can observe that those who haven't used the service yet for a year are the most likely ones to churn.

This trend continues in a downward manner, where the more tenured customers are more likely to stay and retain their use of the service.



Above is the rate of customer churn based on their contract. There are only 3 types of contracts here, Month-to-Month, One Year, Two Year.

From the graph, customers with shorter-term contracts tend to churn while customers with longer-term contracts are expected to stay. This is a common event especially for users who want to try using the product for a month before committing to it and extending their contract.

- d) Based on your exploration, which features do you *hypothesize* will be strong predictors of churn? Explain your reasoning.

Now that exploration has been implemented, based on the features observed, I am hypothesizing that the following features will be strong features:

- Contract - Customers on shorter contracts have less commitment and may be more likely to leave to try other products.
- Tenure in Months - Newer customers may have a higher likelihood of churning, while long-term customers are committed.
- Satisfaction Score - Unhappy customers are more likely to leave.
- Monthly Charge - Higher monthly costs may increase the likelihood of customers switching to competitors and leaving.
- Number of Referrals and Number of Dependents - Customers who actively refer others may be more engaged with the service.

2. Data Preprocessing (20 points)

Goal: *Show that you understand why each preprocessing step is necessary.*

- a) Identify columns to DROP from the analysis (e.g., Customer ID, leaky features like Churn Score, Churn Reason, CLTV). **For each dropped column, explain WHY it should be excluded.**

Next, I believe it is time to remove the columns we don't need.

- Churn Category, Churn Reason, Churn Score - This column is only useful after churn, it doesn't help contextualize why the customer churned
- Customer Status - This will cause the model to have a bias since Customer Status describes if a customer has churned or stayed.
- Customer ID - Unique Identifier
- Lat Long - We already have a latitude and longitude column for geographic information
- Country, State - Since we are in California, we can remove these two columns.

- b) Handle categorical variables using encoding. **Explain your choice between Label Encoding vs One-Hot Encoding and when each is appropriate.**

For this dataset, One-Hot Encoding is better for most categorical variables such as Contract, Internet Type, and Payment Method, as they do not have any ordinal scaling. Label Encoding is then used for binary variables (Yes/no and 1/0), or ordinal columns.

- c) Apply StandardScaler to numerical features. **Explain in plain language why scaling is important for logistic regression.**

Scaling is important in models as it allows for a standardized dataset. All data will be treated as if they are among an equal scale. Large values will not heavily

affect the model, and for lower number values to have an impact on the model.

- d) Report the final feature count. Summarize your preprocessing pipeline in 3-4 sentences that a non-technical manager could understand.

The data was cleaned by removing identifiers and variables that directly describe the churn outcome, which could give the model clues on what to predict.

Categorical information was then converted into separate indicators so that categories such as contract type and payment method could be compared without having any ranks between them.

Numerical variables were then standardized so that features with large values, such as total charges, do not affect the logistic regression model compared with smaller-scale variables such as age. After preprocessing, the dataset contained 2725 predictive features.

3. Model Training (15 points)

Goal: Train a baseline model and assess its generalization.

- a) Train a Logistic Regression model using scikit-learn. Document the hyperparameters you used (e.g., max_iter, solver, C). Why did you choose these values?
- C = 1.0 - The default regularization value. It usually helps the model to stay simple.
 - solver = 'liblinear' - Works good for binary logistic regression and can handle high-dimensional datasets.
 - max_iter = 1000 - Setting to only 1000 so that the model doesn't run for way too long and can converge after 1000 iterations.

- b) Report Accuracy on training, validation, and test sets in a clear table format.

	Dataset	Accuracy
0	Training	0.984615
1	Validation	0.967353
2	Test	0.962385

- c) **Assess model fit:** Is there evidence of overfitting (train >> test) or underfitting (all scores low)? What do the validation results tell you? Explain in 2-3 sentences.

After comparing the training, validation, and test accuracies, we can see that they are extremely high. The relatively small difference between the training and validation/test scores indicates that the model does not show any overfitting nor underfitting for that matter. This indicates that the baseline model generalizes reasonably well.

4. Model Evaluation & Interpretation (25 points)

Goal: Evaluate performance using appropriate metrics and explain what they mean.

- a) Display the Classification Report (Precision, Recall, F1-Score). **In your own words, explain what Precision and Recall mean in the context of churn prediction. Which metric is more important for this business problem and why?**

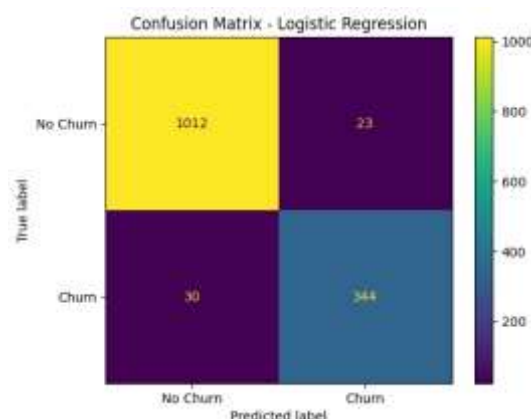
	precision	recall	f1-score	support
No Churn	0.97	0.98	0.97	1035
Churn	0.94	0.92	0.93	374
accuracy			0.96	1409
macro avg	0.95	0.95	0.95	1409
weighted avg	0.96	0.96	0.96	1409

We can proceed with evaluating the model. The most common metrics used for a model like Logistic Regression are the Precision and Recall.

We can see that the Precision of 97% and 94% for No Churn and Churn respectively means that the model can correctly predict those who leave and those who don't. Of all the customers that were predicted to have churned, 94% of those predictions were correct.

For Recall of 92% on Churn, of all the customers who churned, the model guessed 92%. This is the reason Recall is more important, since the primary basis for a high recall is if the model can guess those who churned.

- b) Plot the Confusion Matrix. **Interpret the results: How many customers did we correctly identify as churners (True Positives)? How many churners did we miss (False Negatives)? What is the business cost of each type of error?**

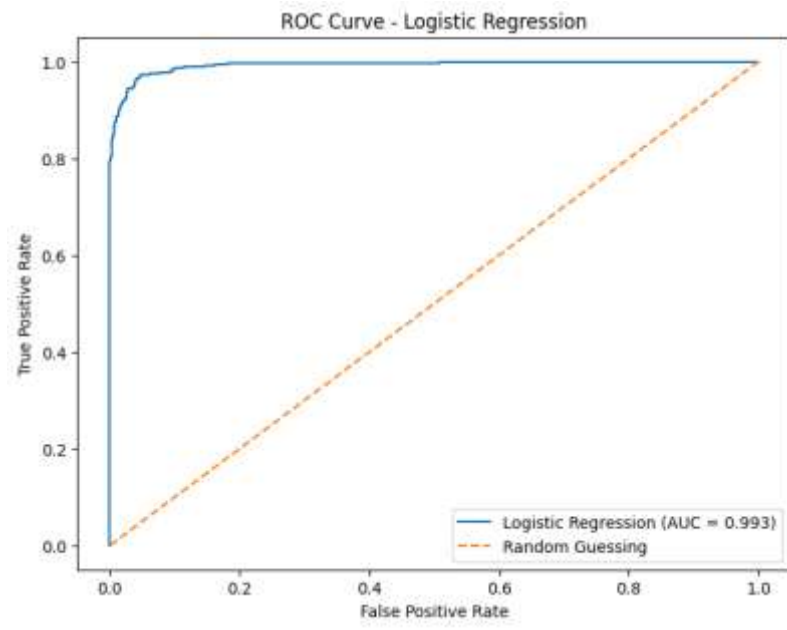


- True Positive (TP)/Top Right: A customer churned and was correctly identified.
- False Negative (FN)/Lower Left: A customer churned but the model predicted that they would stay.
- False Positive (FP)/Top Right: The model predicted churn, but the customer stayed.
- True Negative (TN)/Lower Right: The model predicted no churn, and the customer did stay.

False negatives usually have a higher business cost because these are the customers who churned without being identified for possible retention efforts.

These could be retained customers that would've been beneficial for the telecommunications company, but since they were retained, profits were lost. And if these aren't identified, then there would be an increase in losses in the future.

- c) Calculate ROC-AUC and plot the ROC Curve. **Explain what AUC represents and whether this model is performing better than random guessing.**



The ROC-AUC curve measures if the model can accurately predict, rather than just guess.

With an ROC-AUC value of 0.99 or 99.31%, we can say that the model performs better than just randomly guessing. The closer the AUC reaches 0.50 or 50%, we are essentially guessing the values, but a 1.00 means we can clearly identify rather than guess.

- d) **Executive Summary:** Write a 3-4 sentence summary of model performance that you could present to a non-technical executive.

The logistic regression model achieved an accuracy of 96.24% on the test dataset. This performance indicates significant generalization. The model achieved a ROC-AUC of 99.31%, meaning it can easily distinguish between customers who churn and those who remain.

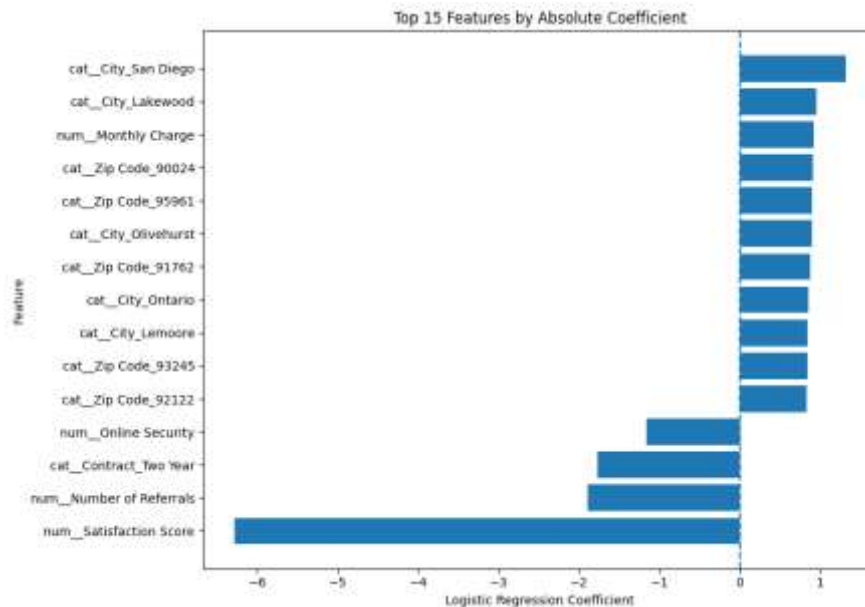
Following that, the confusion matrix shows that the model correctly identified 344 churners, while 30 churners were missed.

For the improvement of the business, improving recall could help identify customers, targeting them for retention campaigns.

5. Feature Importance & Business Insights (20 points)

Goal: Extract actionable insights from model coefficients and translate them into business recommendations.

- a) Extract and display model coefficients sorted by absolute value. Create a visualization (e.g., horizontal bar chart) showing the top 10-15 most important features.



- b) **Interpretation:** Identify the top 5 features that INCREASE churn risk (positive coefficients) and top 5 that DECREASE churn risk (negative coefficients). For each feature, explain in plain language what the coefficient means.

Increase Churn Risk:

- City: San Diego: +1.313 - Customers in San Diego have the highest predicted likelihood of churning.
- City: Lakewood: +0.951 - Customers here have the second highest predicted likelihood of churning.
- Monthly Charge: +0.914 - Higher monthly charges are associated with a higher likelihood of churn.
- Zip Code 90024: +0.903 - Customers in this ZIP code have a higher predicted likelihood of churning.
- City: Olivehurst: +0.900 - Customers here also have a higher predicted likelihood of churning.

Decrease Churn Risk:

- Satisfaction Score: -6.287 - Higher satisfaction score means higher chance of retention and lower likelihood of churn.
- Number of Referrals: -1.895 - Customers with more referrals tend to have a lower likelihood of churning. These customers may be really sold on the service.
- Contract: Two Year: -1.764 - Customers on two-year contracts have a lower likelihood of churn, especially since they are in a committed long-term contract.
- Online Security: -1.163 - Customers with online security are associated with a lower likelihood of churn.

- Zip Code: 92102: -0.744 - Customers in this ZIP code have a lower likelihood of churn.
- c) **Business Validation:** Do these findings make intuitive sense? For example, if "Satisfaction Score" has a negative coefficient, does it make sense that satisfied customers are less likely to churn? Discuss at least 3 features.

Some of these findings do make sense. As hypothesized, some features like Satisfaction Score and Contract Type had an effect on the likelihood of Churn. It makes sense that Satisfaction Score that is high have a negative coefficient, since customers typically would stay with a product they are happy with.

On another note, customers on a long-term contract would be more willing to honor the contract and continue using the service, in this case, for two years. This goes similarly for referrals. Customers who really trust a product will refer it to more people, friends or family.

Lastly, on the other hand, locations do matter. San Diego is a very populated city in California. Meaning, more people will generally tend to churn as opposed to lesser populated cities. The monthly charges also greatly affect a customer's reasoning to stay with a service. They may want to move since they do not have the budget, or rather go for something cheaper.

- d) **Actionable Recommendations:** Based on the coefficients, provide 3 specific, actionable recommendations to the marketing team to reduce churn. Be concrete (e.g., "Offer incentives for Month-to-Month customers to switch to annual contracts").

1. Promotion of longer-term contracts with a reasonable price - The company should encourage a monthly basis of extending the contracts of customers in a monthly contract to a longer-term contract. They can also provide discounts and incentives that may sway the customer to consider, given the monthly charges having a large impact on customer churn.

2. Prioritize customers reporting low satisfaction - To improve satisfaction, the company should follow-up on customers, providing surveys, customer-service interventions, and open feedback when satisfaction scores decline. Knowing the feedback is key to improving customer satisfaction.

3. Consider promotion towards highly populated cities - The company can consider a city like San Diego, with the 2nd largest California population, to expand efforts when it comes to promoting and marketing. They could also focus on customers here to improve customer retention as it is the most impactful one causing churn.

Bonus Questions (Optional)

These questions demonstrate advanced skills. Complete any or all for extra credit.

1. Literature Validation (+5 points)

Research published papers or industry reports on telecom churn prediction. Are

the directions of your coefficients (positive/negative) consistent with findings in the literature? Cite at least one source and discuss any discrepancies.

<https://www.sciencedirect.com/science/article/pii/S2666720726001384>

Fig. 3. Comparison of the most influential features identified by SHAP for IBM Telco and Cell2Cell.

This study above, **MetaLight-ChurnNet: A leakage-safe embedding-enhanced ensemble framework for telecom customer churn prediction**, gathered an ensemble framework in customer churn prediction from IBM Telco and Cell2Cell with Random Forest Classification.

In terms of feature importance, there are similarities along with IBM Telco, such as Month-to-month Contracts and Monthly Charges. Unfortunately, this fictional dataset doesn't have other features featured in these two Telcos, which have a high importance in customer churn prediction.

A discrepancy visible is that IBM Telco had a high score on tenure, but this dataset did not, which was one of the hypothesized important features.

2. Model Improvement (+5 points)

Implement ONE improvement and report updated metrics:

- **Address class imbalance (class_weight='balanced' or SMOTE)**
- Hyperparameter tuning with GridSearchCV
- Compare with Random Forest or XGBoost

Explain: What changed? Did performance improve? Was interpretability affected?

After evaluation, I decided to try evaluate the 'balanced' model since there is a 76-24 ratio class imbalance of No Churn to Churn.

The updated model's performance was compared against the baseline using accuracy, precision, recall, F1-score, and ROC-AUC. Although as we can observe, the original baseline model has better resulting metrics, and the performance didn't improve.

This may be due to the fact that the model places greater emphasis on correctly identifying churners, also known as the minority class. Since we 'balanced' the dataset, we emphasized more on the Churn, which was 26% lesser than the No Churn.

3. Threshold Analysis (+5 points)

The default threshold is 0.5. Experiment with different thresholds and plot Precision vs Recall at various thresholds. **Discuss: From a business perspective, should the company optimize for Precision (avoid wasting resources on customers who won't churn) or Recall (catch as many churners as possible)? What threshold would you recommend and why?**

As we can see, as we increase the threshold, the precision-recall graph converges, but diverges at around 0.45 to 0.5, and slowly becomes opposites of each other.

I took the highest F1-score to identify the best threshold result, and it resulted with 0.45. Lowering the threshold by a bit from 0.5 increased recall and f1-score because more customers were classified as potential churners, but this also increased the false positives and reduced precision.

Based on the observed precision-recall tradeoff, a threshold of 0.45 provides an appropriate balance for that tradeoff.

Evaluation Criteria

Section	Points	Key Evaluation Focus
1. Data Exploration	20	<i>Visualizations, pattern identification, hypothesis</i>
2. Data Preprocessing	20	<i>Justification of decisions, clear explanations</i>
3. Model Training	15	<i>Correct implementation, fit assessment</i>
4. Model Evaluation	25	<i>Metric interpretation, business context</i>

Section	Points	Key Evaluation Focus
5. Feature Importance	20	<i>Insights, recommendations, communication</i>
Required Total	100	
Bonus (Optional)	+15	<i>Literature, improvements, threshold analysis</i>

What We're Looking For

- ✓ Clean, well-organized code with comments
- ✓ Clear explanations of methodology and decisions
- ✓ Proper interpretation of results (not just reporting numbers)
- ✓ Ability to translate technical findings into business language
- ✓ Thoughtful recommendations backed by data

Good luck! We look forward to reviewing your work.